



Faculty of Computers and Artificial Intelligence

Cairo University

CS251: Intro to Software Engineering

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Assignment 1

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1. Document Purpose and Audience

Purpose:

The purpose of this document is to provide a clear description of the requirements of the Personal Budgeting Application. It explains the system features and functionalities.

Audience:

This document is intended for developers, stakeholders, and instructors who are involved in evaluating or developing the system.

2. Introduction

Software Purpose:

The Personal Budgeting Application is designed to help users manage their financial activities by tracking income and expenses.

Software Scope:

The system allows users to:

- Record income and expenses
- Categorize expenses
- Create budgets
- Generate financial reports

It targets students, young adults, and families who want better control over their finances.



1. Definitions and Acronyms :

Definition

Term	Definition
Budget	A plan to control spending
Expense	Money spent by the user
Income	Money received by the user
Category	Type of expense like food or transport
Transaction	Any financial operation
User	Person using the system



Requirements

Functional Requirements:

ID	Requirement Description
FR01	The system shall allow users to create a new account by providing personal information (name, email, password).
FR02	The system shall allow users to log in using their email and password.
FR03	The system shall allow users to log out of their accounts securely.
FR04	The system shall allow users to update their profile information.
FR05	The system shall allow users to reset their password if forgotten.
FR06	The system shall allow users to add income records.
FR07	The system shall allow users to edit income details.
FR08	The system shall allow users to delete income records.
FR09	The system shall categorize income sources (e.g., salary, freelance).
FR10	The system shall allow users to add expenses.
FR11	The system shall allow users to edit expense details.
FR12	The system shall allow users to delete expenses.
FR13	The system shall allow users to categorize expenses (e.g., food, rent, and transport).
FR14	The system shall allow users to create a monthly budget.
FR15	The system shall allow users to set spending limits for each category.
FR16	The system shall notify users when they exceed their budget limits.
FR17	The system shall allow users to update or delete budgets.
FR18	The system shall generate monthly financial reports.
FR19	The system shall display income vs expenses summary.
FR20	The system shall provide visual charts for financial analysis.
FR21	The system shall allow users to filter reports by date.
FR22	The system shall send notifications when the user exceeds the budget.
FR23	The system shall send reminders for upcoming financial goals.
FR24	The system shall allow users to set savings goals.
FR25	The system shall track progress toward savings goals.
FR26	The system shall notify users when a goal is achieved.
FR27	The system shall store all financial data securely.
FR28	The system shall allow users to view their transaction history.



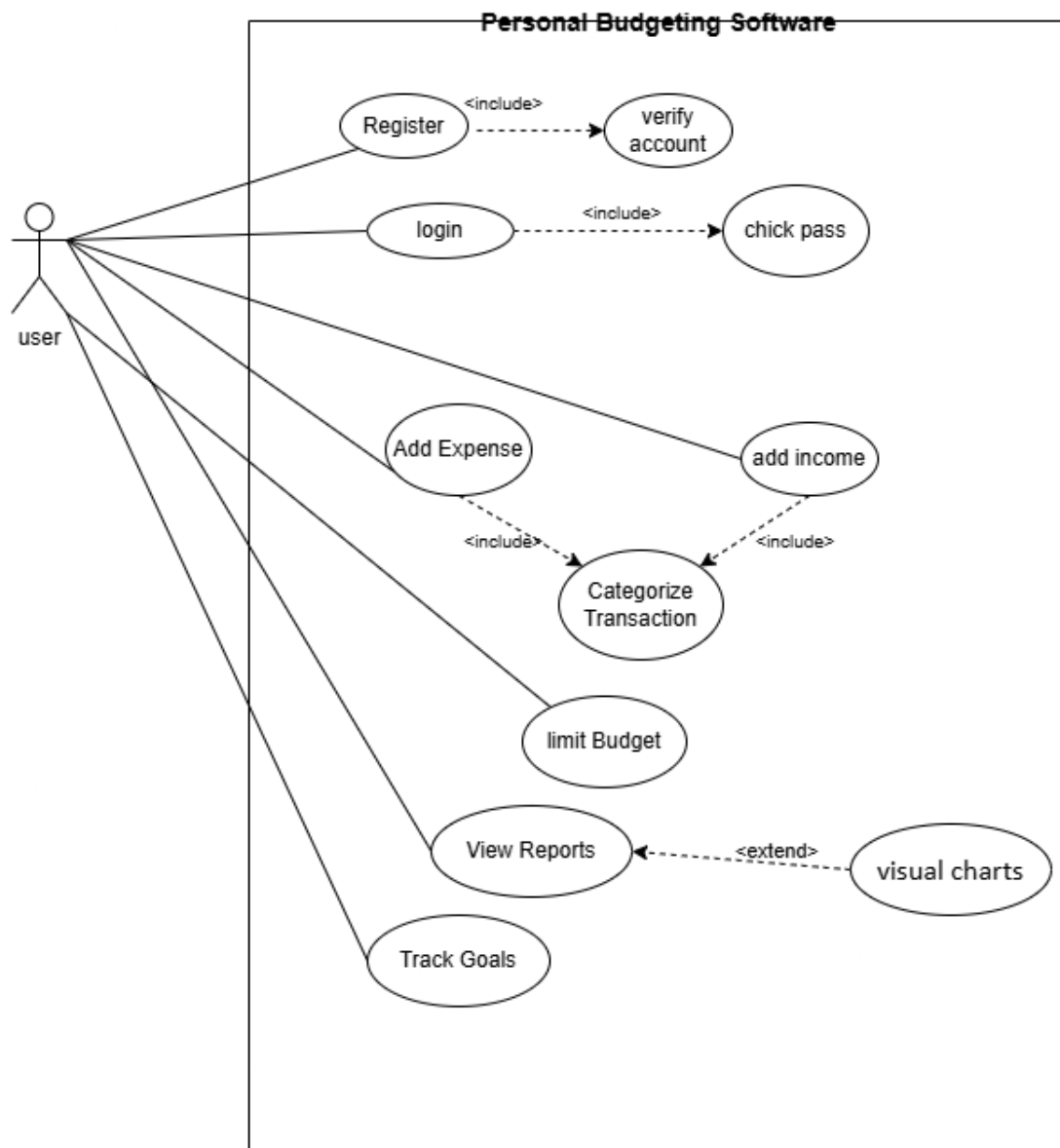
Non-Functional Requirements:

Measure	Details
Performance	• response, financial reports and summaries to user action should be within 2 seconds. • ensure smooth performance on mobile and web platforms.
scalability	• The system should support at least 2,000 concurrent users without performance degradation and support future growth in the number of users. • The system architecture shall allow integration with bank APIs and e-wallets.
Usability	• Provide a simple and user-friendly interface suitable for users aged 18–35. • allow users to add transactions in most 3 steps. • The interface should use clear labels and icons. • support Arabic and English languages. • provide easy and intuitive navigation for users.
Security	• The system should return the pages according to the role of the user (authorize users in login). • ensure that all user data is encrypted during transmission. • Implement secure storage for financial data. • protect user privacy.
Reliability & Availability	• The system should be available 24 hours a day/7 day a week for users. • minimum uptime of 99%. • automatically save user data without loss. • Recover data in case of system failure.
Maintainability	• Designed using modular architecture to allow easy updates. • allow developers to fix bugs and update features. • well-documented for future improvements.
Compatibility	• run on Android and iOS mobile devices. • accessible through modern web browsers (Chrome, Edge, etc.). • support different screen sizes (responsive design).



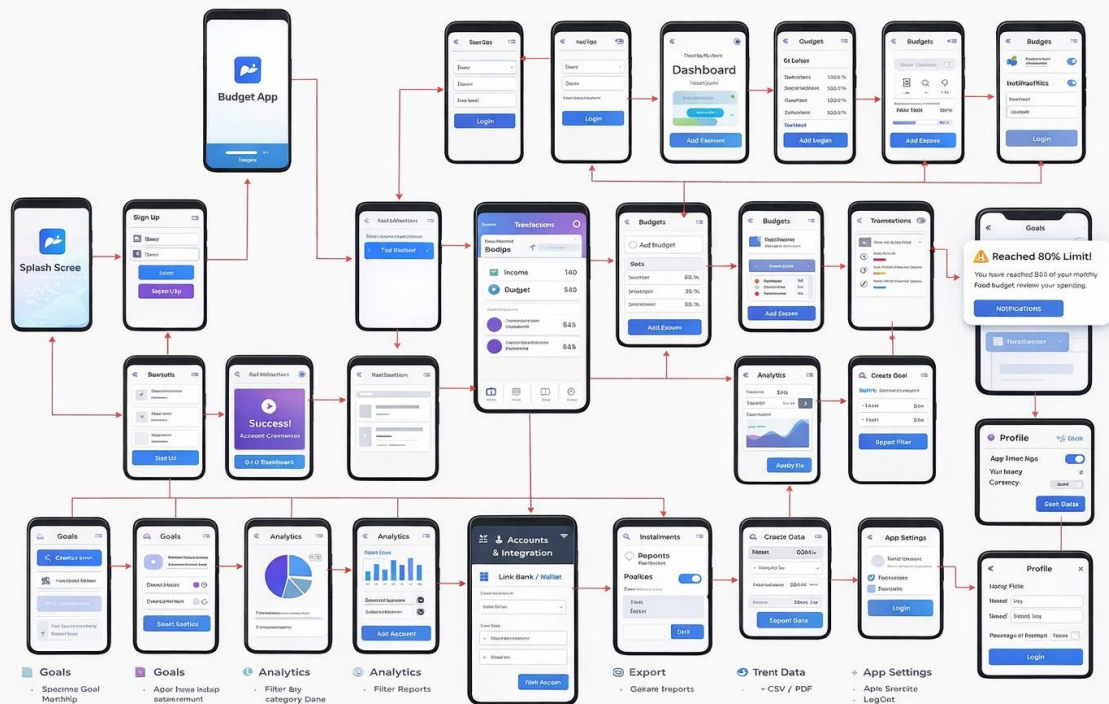
System Model

Use Case Diagram:





System Navigation Map



Enriched User Stories:

● User Story #1

User Story ID	US #1
User Story Name	Account Creation
Actors	Individual User
Description	As a user, I'd like to create a secure account so that my financial data is saved and accessible.
Pre-condition	The user has the application open.
Post condition	User data is saved in the database.



Acceptance Criteria	System validates email format and password strength before creating the account.
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Scenarios

Normal Scenario

Actor Action	System
1. User clicks "Sign Up".	2. System displays registration form.
3. User enters email, name, and password	4. System validates credentials.
5. The user clicks "Submit".	6. System creates an account and sends verification email.

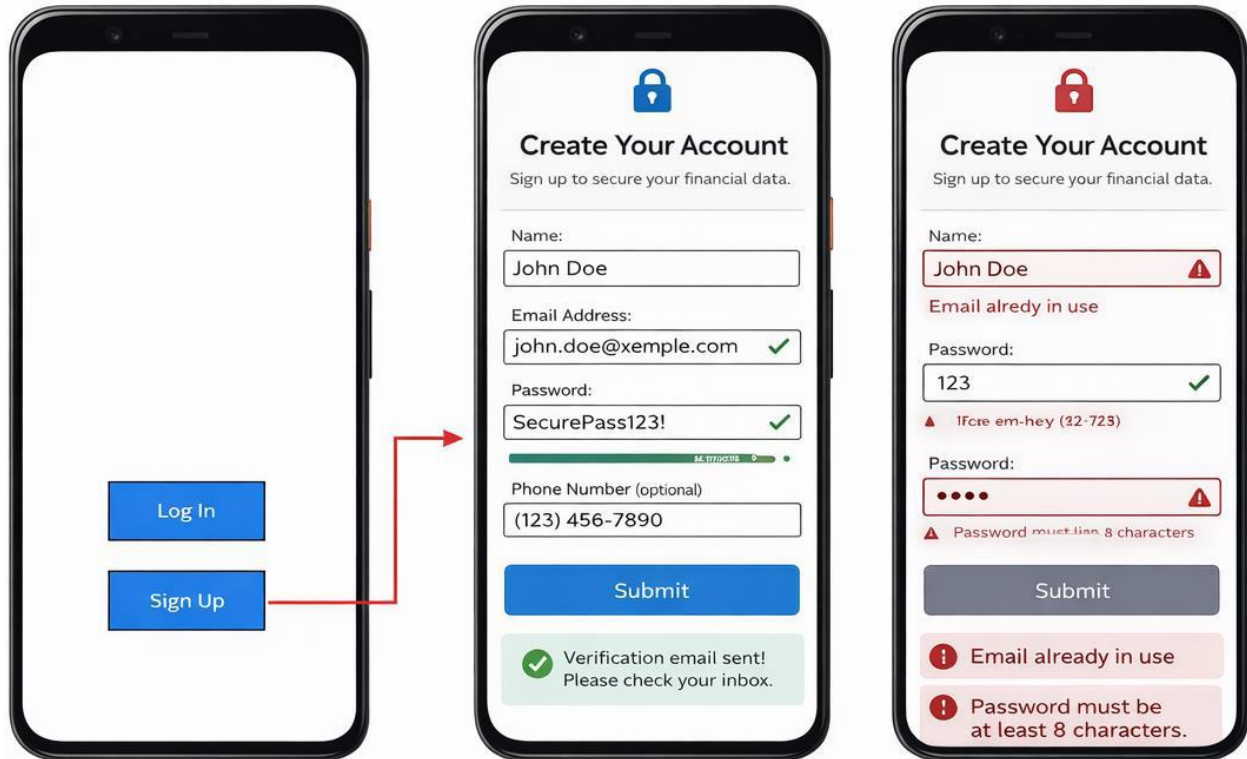
Exceptional Scenario

Actor Action	System
1. User enters an already registered email.	2. System displays "Email already in use" error.
3. User enters a weak password (e.g., "123").	4. System displays "Password must be at least 8 characters."

Screen Design



5.



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Full Name	Text < 50 Characters	Alphabetic characters only.
Email Address	Text < 100 Characters	Must contain "@" and a valid domain (e.g., .com, .edu.eg).
Password	Text < 20 Characters	Minimum 8 characters must include one uppercase and one number.
Sign Up	Button	Trigger validation creates entry in User Database.

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● User Story #2

User Story ID	US #2
User Story Name	Income Entry
Actors	Individual User
Description	As a user, I'd like to log into various sources of income so that I can see my total monthly budget.
Pre-condition	The user is logged in.
Post condition	Total balance is updated.
Acceptance Criteria	Income is added to the "Current Month" view immediately.

Scenarios

Normal Scenario

Actor Action	System
1. User clicks "Add Transaction".	2. System displays entry screen.
3. User selects "Income" type.	4. System shows income categories (Salary, Gift, etc.).
5. User enters 10,000 and clicks "Save".	6. System updates total balance and closes screen.

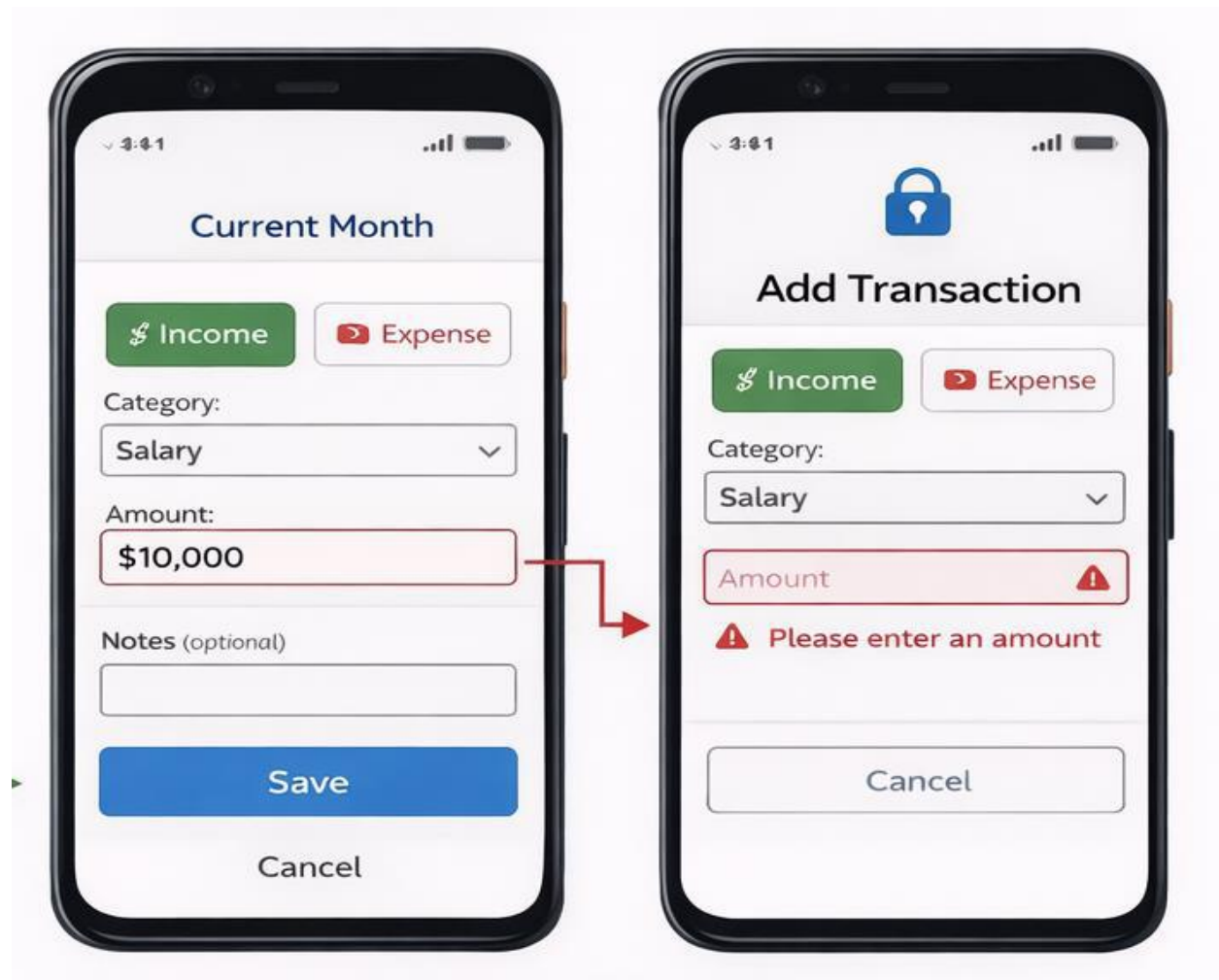
Exceptional Scenario

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Actor Action	System
1. User leaves "Amount" field empty.	
	2. System highlights the field in red and prevents saving.

Screen Design



Data Dictionary:

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Element Label	Type/Length	Data Validation / Business Rule
Income Amount	Decimal (10,2)	Positive value > 0; numeric only.
Source/Note	Text < 100 Characters	Optional; descriptive text for the income.
Date	Date Picker	Defaults to Current Date; cannot be a future date.
Category	Dropdown Selection	Values: Salary, Freelance, Gift, Other.



• User Story #3

User Story ID	US #3
User Story Name	Expense Logging with Categories
Actors	Individual User
Description	As a user, I'd like to categorize my expenses so that I can understand my spending habits.
Pre-condition	User is on the dashboard.
Post condition	Expense is recorded under a specific tag.
Acceptance Criteria	Expense must be linked to a predefined or custom category.

Scenarios

Normal Scenario

Actor Action	System
1. User clicks "Add Expense".	2. System displays expense form.
3. User enters amount and selects "Food".	4. System records transaction under "Food" category.
5. User clicks "Confirm".	6. System subtracts amount from total balance.

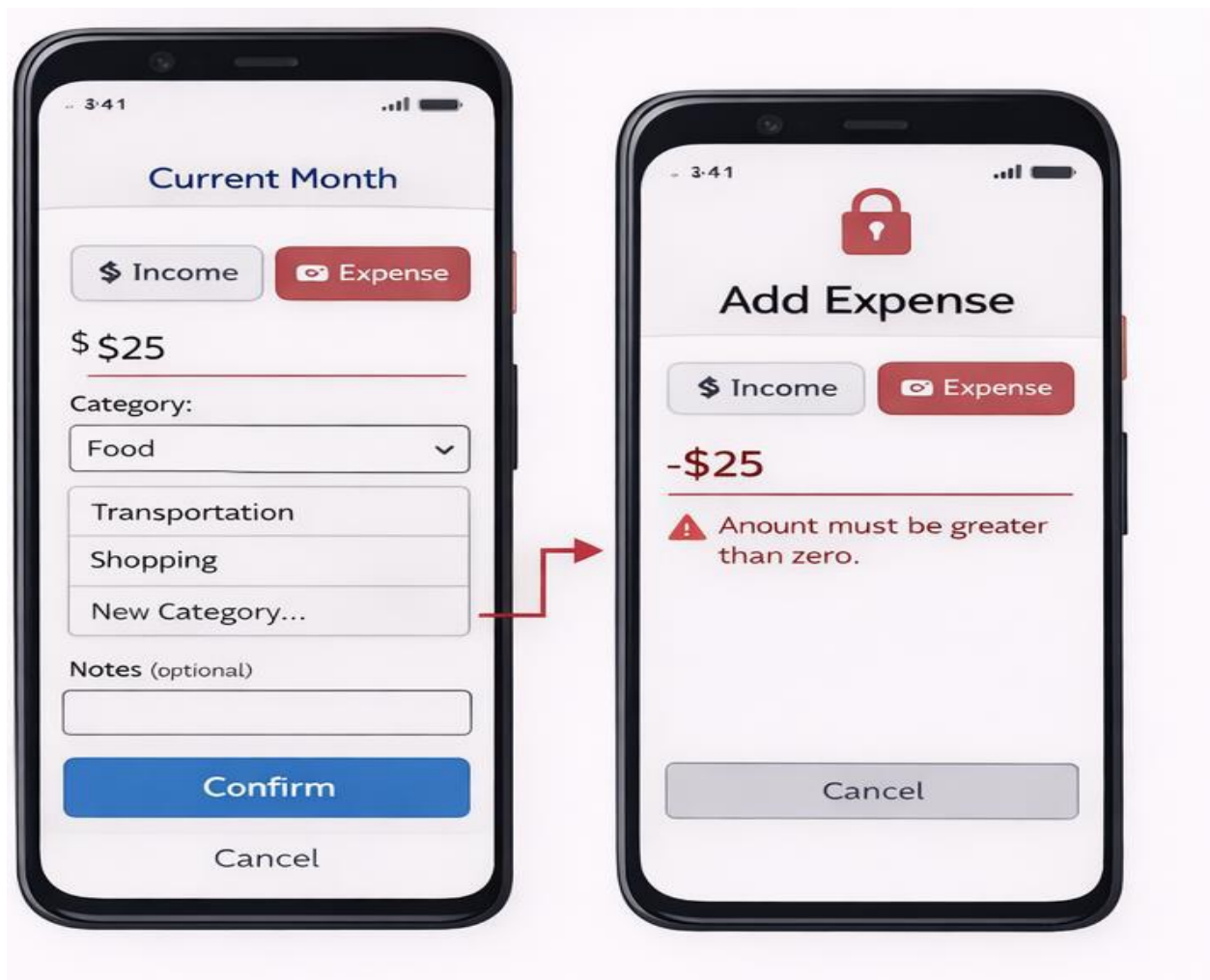
Exceptional Scenario

Actor Action	System
1. User enters a negative amount.	



	2. System displays "Amount must be greater than zero."

Screen Design



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Expense Amount	Decimal (10,2)	Positive value > 0; subtracted from total balance.

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Category	Dropdown Selection	Values: Food, Transport, Rent, Healthcare, etc.
Payment Method	Radio Buttons	Values: Cash, Credit Card, E-Wallet.
Confirm	Buttons	Verifies amount is within available balance or flag deficit.



• User Story #4

User Story ID	US #4
User Story Name	Budget Limit Setup
Actors	Individual User
Description	As a user, I'd like to set spending limits for different categories to avoid overspending.
Pre-condition	Categories are established.
Post condition	A budget limit is stored for the category.
Acceptance Criteria	System displays a progress bar for each budget.

Scenarios

Normal Scenario

Actor Action	System
1. User navigates to "Budgets".	2. System displays current budget list.
3. User selects "Add Budget".	4. System prompts for category and amount.
5. User sets 2,000 for "Transport".	6. System saves limits and displays progress bar.

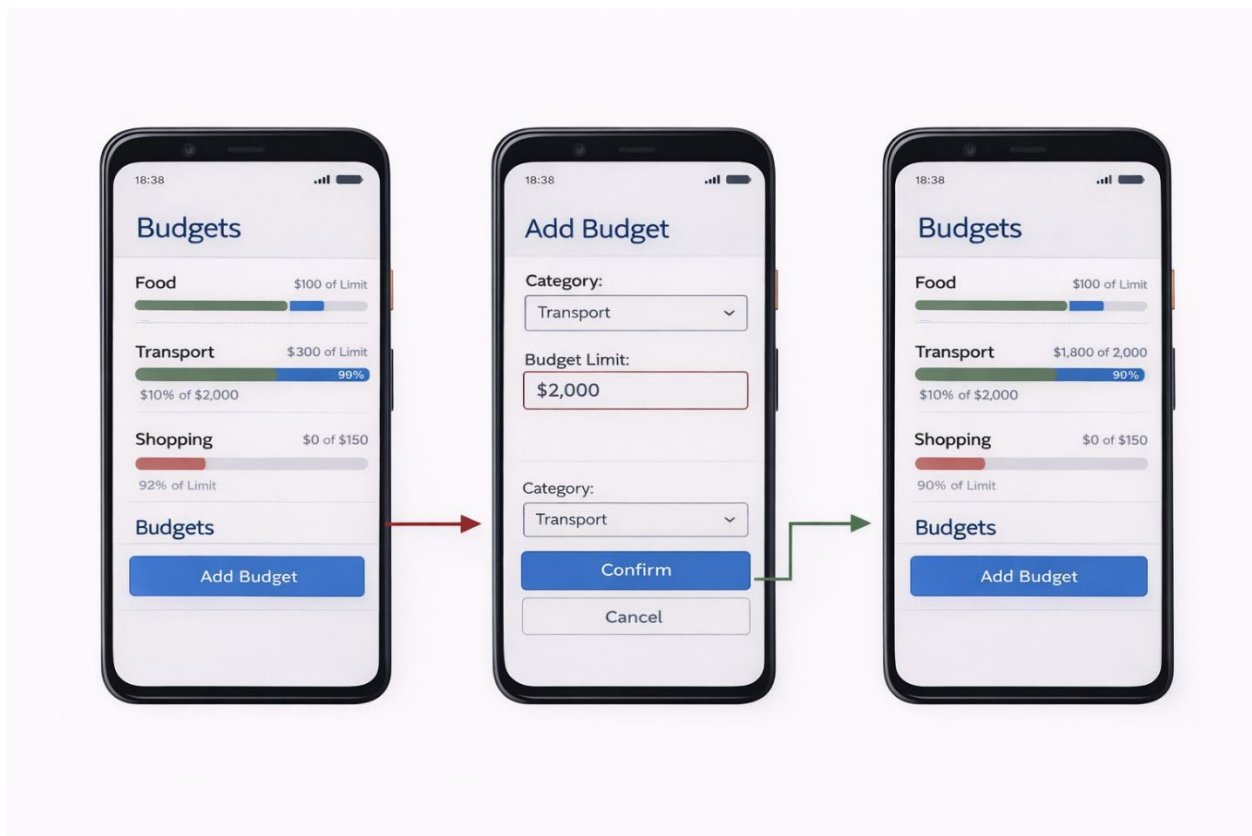
Exceptional Scenario

Actor Action	System
1. User sets a budget of 0.	



	2. System prompts "Budget must be a positive value."

Screen Design



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Budget Limit	Integer < 6 digits	Total allowed spending for a specific category.
Category Name	Dropdown Selection	Linked to existing expense categories from US #3.
Time Period	Selection	Monthly (default) or Weekly.

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Save Budget	Button	Overwrites existing budget if category is already set.
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• User Story #5

User Story ID	US #5
User Story Name	Budget Notifications
Actors	Individual User
Description	As a user, I'd like to receive notifications when I reach 80% of my budget limit.
Pre-condition	A budget has been set in Budget Limit Setup
Post condition	A notification is delivered to the device.
Acceptance Criteria	Alert triggers automatically upon transaction entry.

Scenarios

Normal Scenario

Actor Action	System
1. User enters a 500 EGP expense.	2. System calculates total category spending.
3. Spending reaches 80% of limit	4. System triggers a push notification warning.

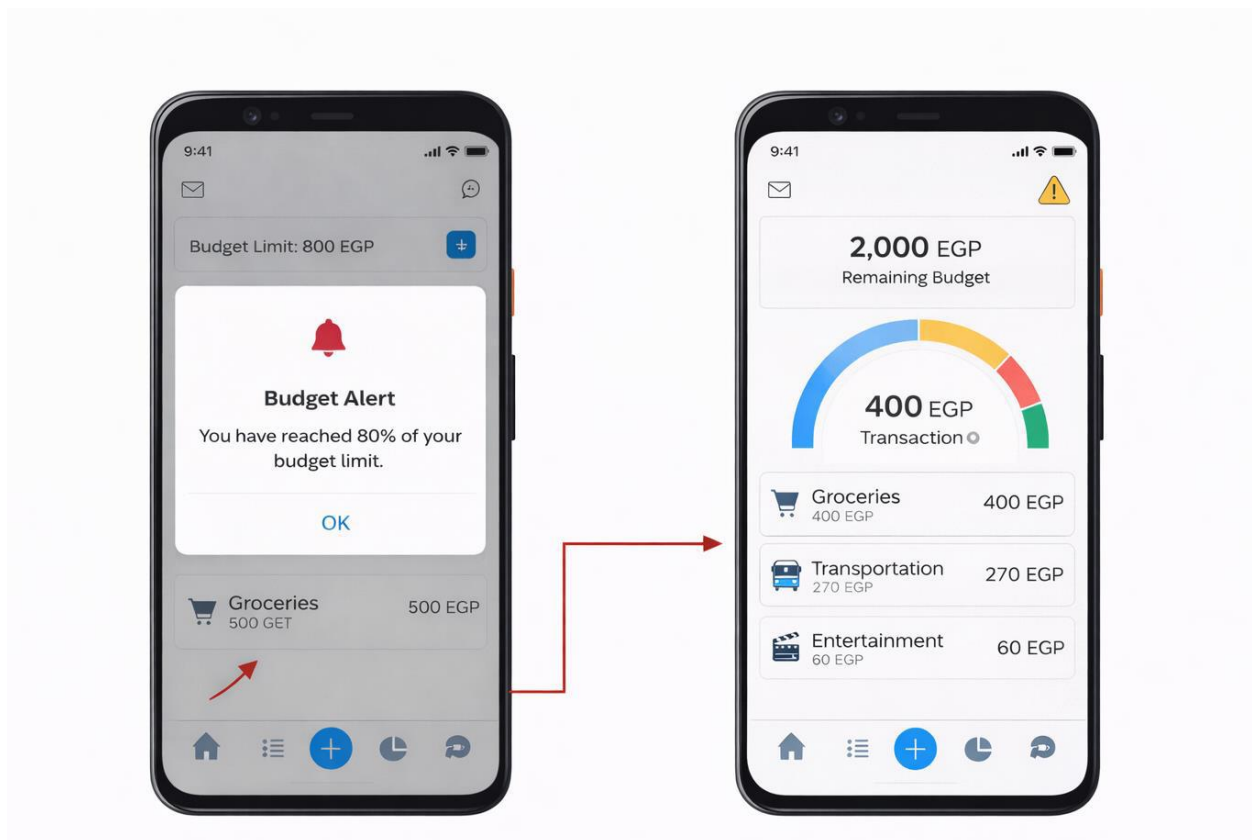
Exceptional Scenario

Actor Action	System
1. User has notifications disabled.	



	2. System displays a warning icon on the app dashboard.

Screen Design



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Threshold %	Integer (1-100)	Percentage of budget used before alert (Default: 80%).
Alert Toggle	Boolean (Switch)	If OFF, push notifications are suppressed.
Notification Sound	Selection	Choose from system default tones.

● User Story #6



User Story ID	US #6
User Story Name	Long-term Savings Goals
Actors	Individual User
Description	As a user, I'd like to set specific savings goals to track my progress.
Pre-condition	The user has a profile.
Post condition	A goal is created with a target date.
Acceptance Criteria	The system must calculate monthly savings needed.

Scenarios

Normal Scenario

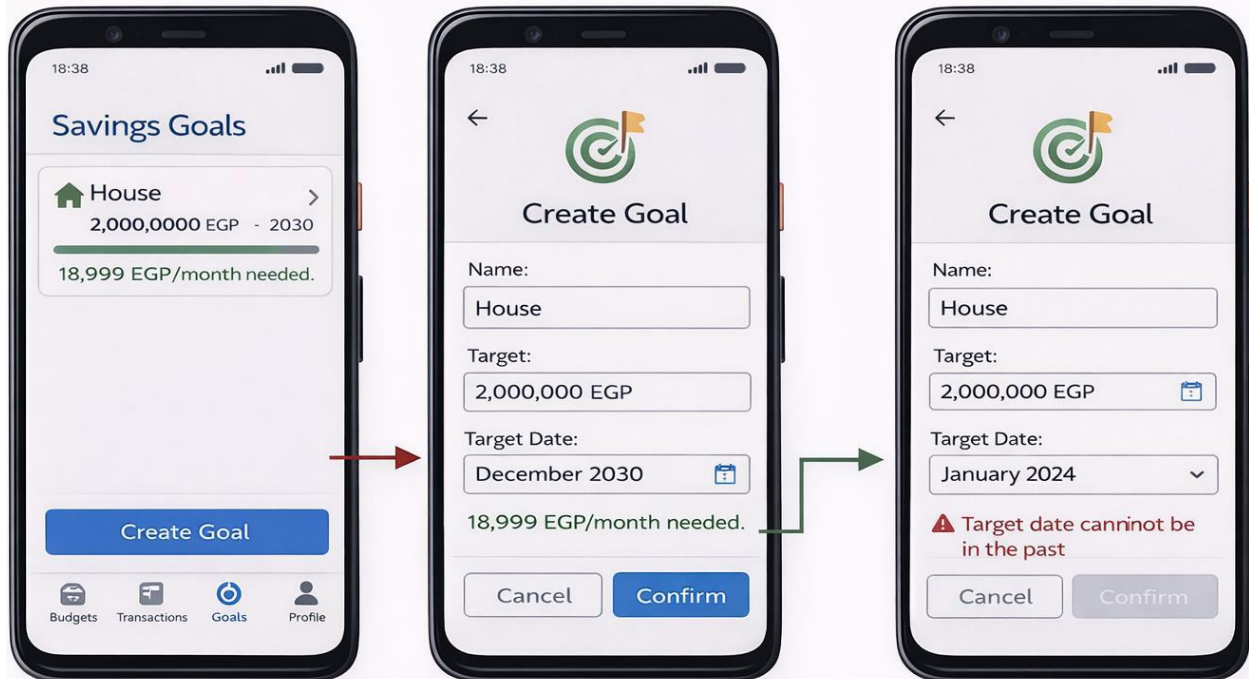
Actor Action	System
1. User clicks "Create Goal".	2. System asks for Name, Target, and Date.
3. User enters "House", "2M", "Dec 2030".	4. System saves goal.
5. User views goals.	6. System shows "18,999 EGP/month needed."

Exceptional Scenario

Actor Action	System
1. User sets date to the past month.	2. System displays "Target date cannot be in the past."



Screen Design



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Goal Title	Text < 30 Characters	e.g., "Umrah Trip," "New Laptop."
Target Amount	Integer < 8 digits	Total savings objective in EGP.
Target Date	Date Picker	Must be a future date.
Priority Level	Dropdown	High, Medium, Low (used for sorting).

• User Story #7



User Story ID	US #7
User Story Name	External Account Integration
Actors	Individual User
Description	As a user, I'd like to sync my e-wallets or bank accounts for automated logging.
Pre-condition	Valid bank credentials or API access.
Post condition	Transactions are auto populated.
Acceptance Criteria	Data must be encrypted during transfer.

Scenarios

Normal Scenario

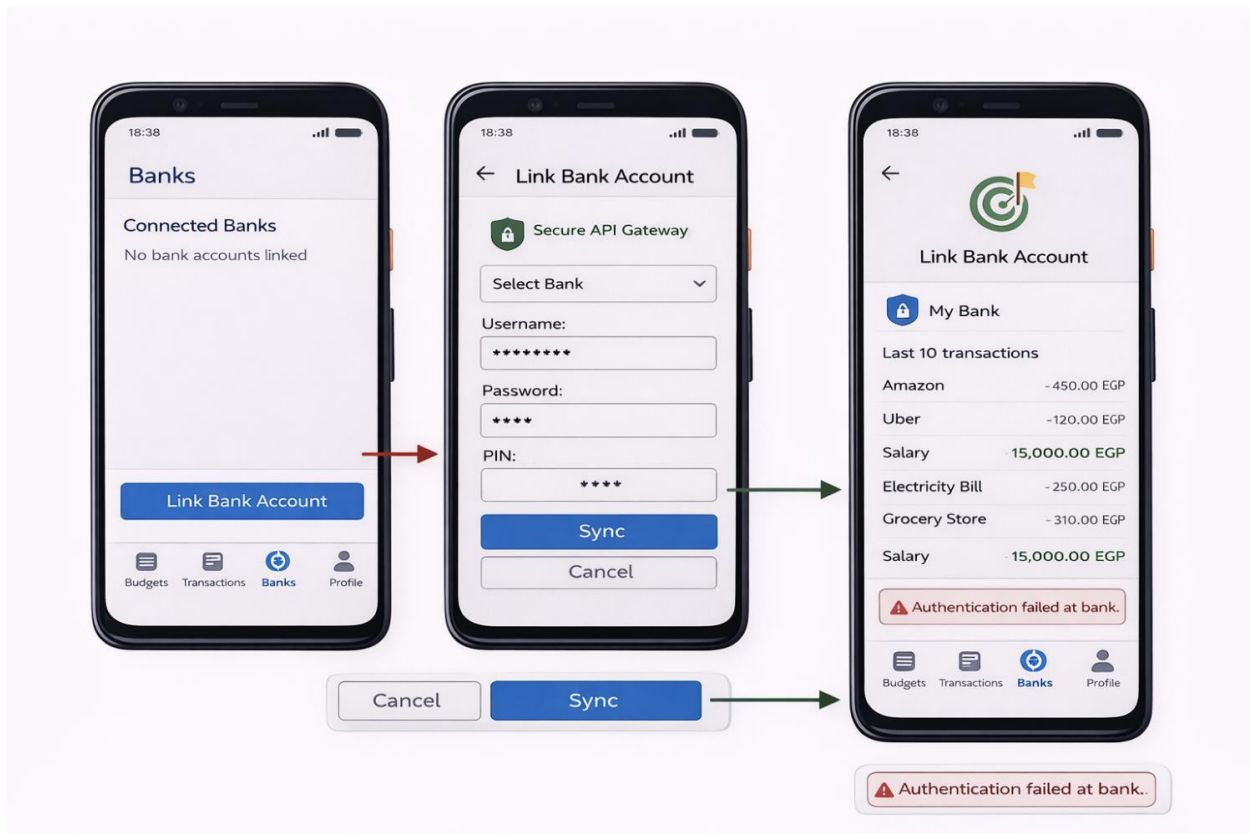
Actor Action	System
1. User selects "Link Bank Account".	2. System opens secure API gateway.
3. User enters bank credentials.	4. System authenticates with the bank.
5. User clicks "Sync".	6. System imports the last 10 transactions.

Exceptional Scenario

Actor Action	System
1. User enters incorrect bank PIN.	
	2. System displays "Authentication failed at bank."



Screen Design



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Provider Name	Dropdown	Selection of Egyptian Banks (CIB, QNB) or E-wallets (Vodafone Cash).
API Token/Login	Encrypted String	Secure authentication via third-party gateway.
Sync Frequency	Selection	Daily, Weekly, or Manual.
Last Sync	Text (Read-Only)	Timestamp of the most recent data pull.

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• User Story #8

User Story ID	US #8
User Story Name	Graphical Data Insights
Actors	Individual User
Description	As a user, I'd like to view charts of my spending patterns to identify where my money goes.
Pre-condition	At least 5 transactions exist.
Post condition	Charts are rendered.
Acceptance Criteria	Charts must toggle between Pie and Bar view.

Scenarios

Normal Scenario

Actor Action	System
1. User clicks "Analytics".	2. System compiles data into a Pie chart.
3. User taps on "Rent" segment.	4. System shows total amount spent on Rent.
5.	6.

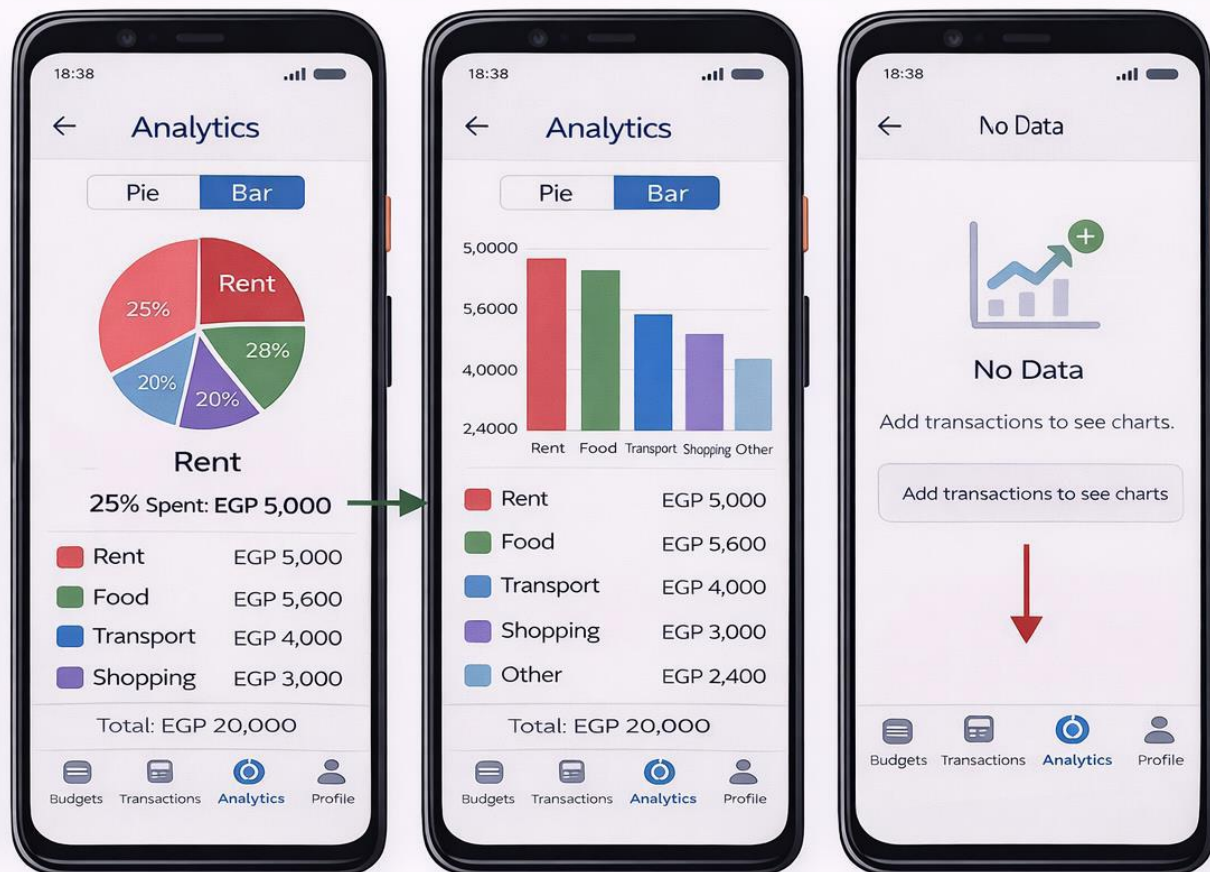
Exceptional Scenario

Actor Action	System
1. User clicks analytics with 0 data.	



	2. System displays “Add transactions to see charts.”

Screen Design



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Report Range	Date Range	Start and End dates for data visualization.
Chart Type	Toggle	Switch between Pie Chart (Percentage) and Bar Chart (Comparison).

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Category Filter	Multi-Select	Filter report to show specific categories only.
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• User Story #9

User Story ID	US #9
User Story Name	Installment Reminders
Actors	Individual User
Description	As a user, I'd like to track my installments so I can plan monthly fixed costs.
Pre-condition	User has an upcoming bill.
Post condition	A recurring reminder is set.
Acceptance Criteria	Reminder alerts user 24 hours before the due date.

Scenarios

Normal Scenario

Actor Action	System
1. User adds "Monthly Internet Bill".	2. System asks for amount and due date.
3. User sets "600 EGP" on the "1st".	4. System creates recurring event.
5. Date is 31st of the month.	6. System sends "Bill due tomorrow" alert.

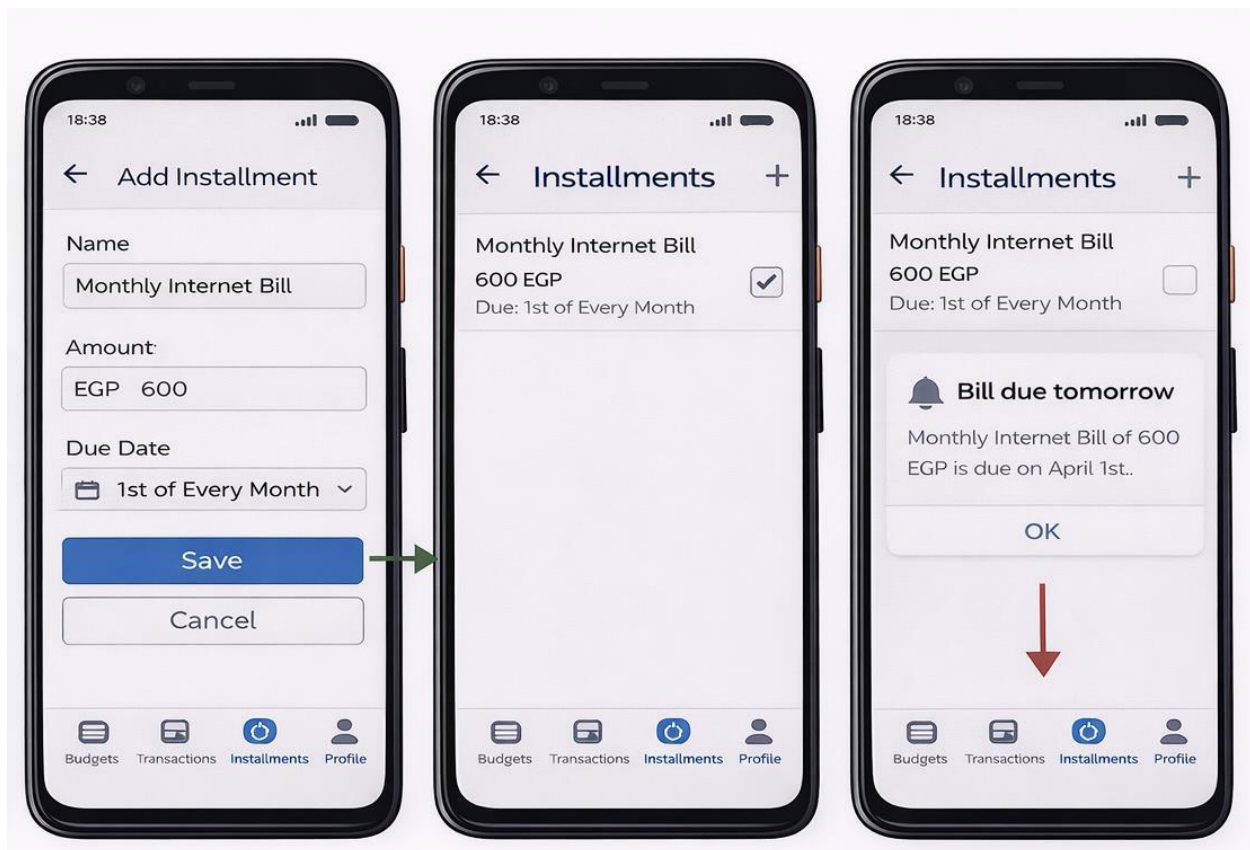
Exceptional Scenario

Actor Action	System
1. User marks bill as "Paid" twice.	



	2. System ignores second click to avoid duplicate entry.

Screen Design



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Debt Name	Text < 50 Characters	e.g., "Car Loan," "Friend X."
Monthly Due	Decimal (10,2)	Recurring amount deducted automatically in projection.
Remaining Debt	Decimal (10,2)	(Total Debt - Payments made).

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Due Day	Integer (1-31)	The day of the month the payment is required.



● User Story #10

User Story ID	US #10
User Story Name	Religious Compliance Filtering
Actors	Individual User
Description	As a user, I'd like to flag transactions as non-halal (interest-based) to maintain my values.
Pre-condition	Features are enabled in settings.
Post condition	Interest-based money is segregated in reports.
Acceptance Criteria	A "Halal Status" icon appears next to transactions.

Scenarios

Normal Scenario

Actor Action	System
1. User selects a "Bank Interest" entry.	2. System provides a "Mark as Riba" toggle.
3. User toggles the switch to "On".	4. System flags the entry with a specific icon.

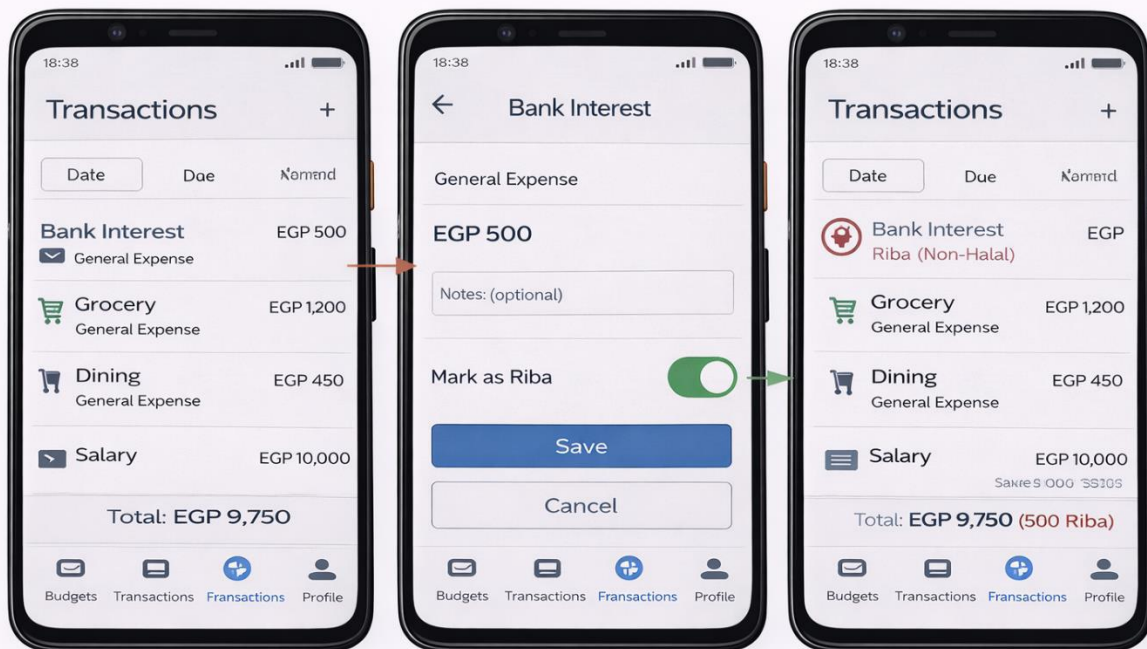
Exceptional Scenario

Actor Action	System
1. User tries to flag "Grocery" as Riba.	



	2. System saves it but shows a "General Expense" warning.

Screen Design



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Halal/Interest Flag	Boolean (Toggle)	Manual flag to identify "Riba" or non-permissible interest.
Interest Type	Dropdown	Values: Bank Interest, Late Fee, Loan Interest.
Status Icon	Image/Icon	Visual marker (e.g., Green/Red dot) next to transaction.

● User Story #11



User Story ID	US #11
User Story Name	Currency Conversion Entry
Actors	Individual User
Description	As a user, I'd like to log expenses in foreign currencies during travel.
Pre-condition	Internet connection for exchange rates.
Post condition	Expense is converted and saved in EGP.
Acceptance Criteria	Conversion uses the current market rate.

Scenarios**Normal Scenario**

Actor Action	System
1. User opens "Add Expense".	2. System shows default currency (EGP).
3. User changes currency to "USD".	4. System fetches current USD/EGP rate.
5. User enters "10".	6. System saves as "510 EGP" (approx.).

Exceptional Scenario

Actor Action	System
1. User is offline.	
	2. System uses the last saved exchange rate.

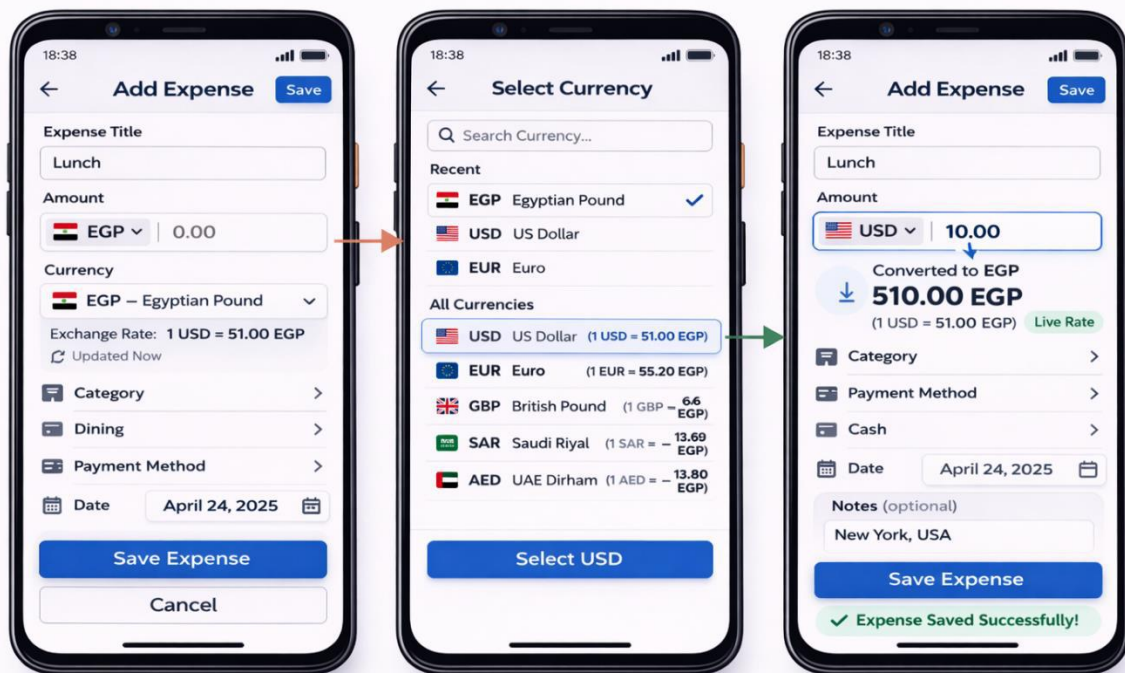
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Screen Design



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Base Currency	Text (3 Chars)	Defaults to "EGP" based on regional settings.
Foreign Amount	Decimal (10,2)	Original transaction amount in foreign currency.
Exchange Rate	Decimal (6,4)	Fetches daily from central bank API.
Converted Total	Decimal (Read-Only)	Result of (Foreign Amount * Exchange Rate).

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• User Story #12

User Story ID	US #12
User Story Name	PDF/Excel Data Export
Actors	Individual User
Description	As a user, I'd like to export my monthly summary to share it with family.
Pre-condition	Data exists for the selected month.
Post condition	A PDF file is generated.
Acceptance Criteria	Exports must include all categories and total savings.

Scenarios

Normal Scenario

Actor Action	System
1. User navigates to "Export".	2. System asks for Month and Format.
3. User selects "March" and "PDF".	4. System generates and downloads the file.

Exceptional Scenario

Actor Action	System
1. User selects a month with no data.	



	2. System displays "Nothing to export for this period."

Screen Design



Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Format Selection	Radio Buttons	Values: PDF (Read-only), CSV (Excel-compatible).
Content Scope	Checkboxes	Values: Summary Only, Full Transaction List, Charts.
Export Button	Button	Triggers file generation and local download.

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Tools

[Draw.io](https://draw.io) For Use Case

[mockflow](https://mockflow.com) For System Navigation Map

Ownership Report

student	Items he/she created
Yasmin Mohamed Nabil	• Non-Functional Requirements • Use Case Diagram
Lojain Mohamed Abu Elhassan	• Functional Requirements
Mario Nasser Fawzy	• User Stories
Mahmoud Yasser Mahmoud	• Intro • Definitions